

Day0400 -- IoTs Platform and SCADA System			
เจ้าของงาน ID:		Name:	
คู่ทดลอง ID:		Name:	

- Download >>
https://drive.google.com/drive/folders/1gLpJZLV8fJLuK5wv41XhhycfM2eyHfGh?usp=drive_link
- ทำการทดลอง เก็บข้อมูล ลองคนต่อกลุ่ม แล้วแยกกันเติมรายละเอียดรายงานให้เรียบร้อย
- กำหนดชื่อไฟล์ตามรูปแบบ “B3601234 - Wichai Srisuruk – Day0400 IoTs and SCADA.pdf”
- ส่งแบบ pdf รอบโครงร่าง ก่อน 20260815-1700 ที่ <https://forms.gle/MWPNSayVonMBjEdn8>
- ส่งแบบ pdf รอบสมบูรณ์ ก่อน 20260830-0600 ที่ <https://forms.gle/MWPNSayVonMBjEdn8>

Experiment 1/3: ESP32C6 Microcontroller

Test-01. Bluetooth

Test-02. HTTP Webservice

Test-03. MQTT

Test-04. CoAP

Test-05. ESP-Now

Test-06. Zigbee

Test-07. Thread / Matter

Test-08. LoRa WAN

Test-09. Modbus RTU – Master, Slave

Test-10. Modbus TCP – Server, Client

Experiment 2/3: ESPHome

How ESPHome works

Building your own smart home device from scratch can feel daunting, particularly if you're not familiar with coding. Not with ESPHome. Its intuitive Device Builder replaces code with simple visual choices for a hassle-free setup that brings you straight into building without touching a command line. And for experienced users who prefer writing code manually, YAML is always there if you need it. Every device works out of the box with Home Assistant, too – locally, privately, and securely.

Step 1: Choose & configure: Select your board and components using the visual Device Builder or a YAML file. Pick sensors from the catalog and set your options without writing code.

Step 2: Build & install: ESPHome installs your configuration to your device straight from your browser, or wirelessly over the air.

Step 3: Connect & control: Your new device automatically pops up in Home Assistant over your local network: no cloud accounts or external servers required.

Read More

- <https://esphome.io/>

Experiment 3/3: ESP32C6 and AI

แนวทางการนำ LLM มาใช้บน ESP32-C6

Read More

- Running a 28.9M parameter LLM on a microcontroller
<https://github.com/slvDev/esp32-ai>
- <https://www.facebook.com/photo?fbid=1691033469697857&set=pb.100063736613709.-2207520000>